Handiko Gesang Anugrah Sejati

RETAIL ALGORITHMIC ANALYST & TRADER

Portfolio: handiko.github.io/algo/ **Source Code:** github.com/handiko **Focus:** Quantitative Engineering

EXECUTIVE PROFILE

Quantitative-focused Algorithmic Trader and Systems Developer with a sophisticated understanding of market microstructure across Cryptocurrencies, Foreign Exchange (Forex), and Equities. Specialized in designing, building, and benchmarking high-fidelity automated execution systems and Expert Advisors (EAs) within the MQL5 ecosystem. Broad domain expertise emphasizes structural order book dynamics, stochastic liquidity modeling, stringent multi-tier risk mitigation frameworks, and algorithmic market making.

Proven capability in translating empirical statistical anomalies and probabilistic frameworks (e.g., Markov-Chain models, Monte Carlo optimizations) into production-ready software solutions. Transitioning core technical capabilities in market mechanics into institutional Crypto Business Operations and Liquidity Management to structurally optimize exchange micro-liquidity, control toxic order flow, and maximize capital execution efficiency.

CORE TECHNICAL COMPETENCIES

EXECUTION SYSTEMS

- MQL5 Expert Advisors
- MetaTrader 5 Architecture
- Algorithmic Trade Automation
- MQL5 Backtesting Suites
- Low-Latency Execution
- Pine Script v5

QUANTITATIVE ANALYTICS

- Order Book Dynamics
- Market Microstructure
- Liquidity Aggregation
- Markov-Chain Analysis
- Time-Series Engineering
- Stochastic Modeling

RISK & PORTFOLIO MANAGEMENT

- Monte Carlo Simulations
- Optimal Frontier Modeling
- P&L Optimization
- Value-at-Risk (VaR)
- Inventory Risk Management
- Drawdown Mitigation

IN-DEPTH ENGINEERING PORTFOLIO

Category I: Market Making, Liquidity Architecture & Exchange Simulation

Avellaneda-Stoikov Market Maker Simulator

View Project →

An interactive, high-fidelity deployment of the seminal Avellaneda-Stoikov (2008) framework modeling stochastic control problems for high-frequency liquidity providers. The system solves the continuous-time Hamilton-Jacobi-Bellman equation to dynamically compute inventory-adjusted optimal bid-ask reservation spreads. It explicitly models limit order execution probabilities as a Poisson intensity function dependent on distance to mid-price, providing structural validation for managing skew and mitigating multi-asset inventory risks under adverse selection.

STOCHASTIC CONTROL & SIMULATION

USDT/IDR Monitoring Terminal & Liquidity Dashboard

View Project →

A low-latency monitoring infrastructure designed as a "Terminal-as-a-Service" UI for the USDT/IDR stablecoin pair across regional Indonesian digital asset venues. Focuses heavily on deep-tier order book aggregation, microsecond order book tracking, and slip-page simulation. Developed to provide operators with comprehensive cross-venue order book depth transparency, minimizing latency variance and evaluating execution pathing for large block transactions.

LIQUIDITY INFRASTRUCTURE

Treasury Monitoring Terminal: Smart Rebalancing System

View Project →

A specialized exchange operations simulation application built to validate multi-venue capital rebalancing policies. Simulates automated exchange treasury heuristics designed to optimize reserve capital matching, hot/cold wallet distribution ratios, and operational liquidity buffers. Focuses on minimizing the frictional costs of cross-venue currency rebalancing and preventing structural inventory-outage events under asymmetric order flow.

EXCHANGE OPERATIONS

Category II: Quantitative Portfolio Optimization & Strategic Performance Analytics

Multi-Criteria Optimal Portfolio Allocation Engine

View Project →

A programmatic Python-based capital allocation framework executing multi-thousand iteration Monte Carlo simulations to extract mathematically optimal asset weights. Moving aggressively beyond simplistic Mean-Variance frameworks, the model tracks multi-criteria metrics including Sortino ratios, Maximum Drawdown, and tail-risk indicators, establishing an adaptive efficient frontier curve tailored to structural risk thresholds.

PORTFOLIO OPTIMIZATION

Quantitative Portfolio Performance Simulation and Analysis

View Project →

A comprehensive historical backtesting and statistical evaluation engine designed to rigorously analyze custom, data-driven optimal portfolios against major benchmark equity indices. Tracks and charts structural tracking errors, rolling annualized beta profiles, idiosyncratic alpha generation, and comprehensive downside variance metrics over long historical horizons.

PERFORMANCE BACKTESTING

Category III: Systematic Strategy Engineering & Expert Advisor Optimization

Markov-Chain Conditioned Strategy Architecture

View Project →

Production-level implementation of a live systematic trading strategy parameterized directly by discrete-time Markov Chain probability matrices. The model filters alpha entry signals based on state-transition paths to dynamically alter risk settings, position size, and holding duration. Features rigorous transaction cost adjustments to mirror real-world execution.

SYSTEMATIC ALPHA GENERATION

Structural MQL5 Strategy Optimization Framework

View Project →

An architecture-focused engineering case study detailing a step-by-step framework for isolating and eliminating structural bugs, execution latencies, and look-ahead biases from legacy MQL5 Expert Advisors. Introduces advanced object-oriented state tracking, localized volatility-adjusted trade exits, and rigorous multi-currency stress testing modules.

MQL5 CORE ENGINEERING

Dual-Indicator Mean Reversion Backtesting Pipeline (200-SMA & 2-RSI)

View Project →

A custom-built Python backtesting framework testing low-duration mean-reversion anomalies. Utilizes a long-term 200-period Simple Moving Average as a structural trend filter, paired with an ultra-short 2-period Relative Strength Index to time extreme localized overbought/oversold impulses, outputting exact execution logs and performance metrics.

TECHNICAL INDICATOR ENGINEERING

Multi-Asset Portfolio Strategy Cross-Backtester

View Project →

An event-driven portfolio testing engine tracking technical indicator confluences across massive, heterogeneous baskets of equity and commodity tickers. Simulates concurrent cross-asset asset allocations, capital compounding, variable margin utilization, and real-world slippage layers to isolate systemic correlation vectors during market stress.

EVENT-DRIVEN BACKTESTING

RSI-2 Systematic Strategy (Pine Script v5 Implementation)

View Project →

A robust script built for TradingView to programmatically plot and execute trend-following mean reversion frameworks. Integrates Exponential Moving Average filters with high-frequency RSI trigger levels. Configured for instantaneous webhook transmission to communicate seamlessly with retail brokerage APIs and execution bridges.

PINE SCRIPT AUTOMATION

Category I/V: Statistical Risk Modeling & Quantitative Data Infrastructure

Discrete-Time Markov Chain Forex Regime Classifier

View Project →

Models daily close-to-close returns across major FX pairs as discrete-time Markov chains. Resolves multi-state stationary distribution matrices to establish whether sequential positive or negative price days possess historical statistical significance, isolating hidden micro-structural imbalances from white noise.

STOCHASTIC ANALYSIS

Markovian Financial Volatility & Tail-Risk Predictor

View Project →

An advanced risk management model applying multi-state Markov chains to classify asset volatility shifts. By computing the empirical transition probability matrix between low, moderate, and extreme tail-risk regimes, the system provides dynamic, adaptive inputs for modifying institutional Value-at-Risk parameters.

RISK ANALYTICS

Comprehensive Market Seasonality Analytics Engine

View Project →

An automated quantitative script extracting multi-decade historical time series across any global instrument mapped on Yahoo Finance. Cleans and aggregates sparse matrices into statistical calendar and cyclical seasonality dashboards, evaluating long-range structural time-series configurations.

TIME-SERIES MINING

Monthly Structural Seasonality Backtesting Tool

View Project →

A strategy model exploring structural anomalies associated with calendar-month cycles. Simulates systemic long/short actions optimized based on historical multi-decade recurring probability matrices, calculating specific performance thresholds across asset classes.

CYCLICAL ANALYTICS

High-Fidelity Price Data ETL & MT5 Format Converter

View Project →

A robust data pipeline utility engineered to download public market data arrays and cleanly structure, parse, and encode them into native MetaTrader 5 offline history formats (.hst / .csv), maintaining strict corporate timezone alignments for accurate local optimization.

DATA ENGINEERING